Passive RF-Silent Drone Detection via Neuromorphic Vision and Acoustic Sensor Fusion of the Propeller Blade-Pass Frequency

Giray Yillikci
Independent Researcher
gyillikci@gmail.com

İsa Özyıldırım
Independent Researcher

***Abstract*—We investigate passive, RF-silent drone detection by exploiting the fact that a spinning propeller emits a characteristic blade-pass frequency (BPF) simultaneously in two physical channels: optical flicker, sensed by a neuromorphic event camera, and acoustic pressure, sensed by a microphone array. To our knowledge no published work fuses these two modalities for counter-UAS purposes. We present a working prototype that (i) extracts per-pixel flicker frequency from a Prophesee EVK4-HD event camera and isolates propeller regions via connected-component spatial clustering and temporal tracking, (ii) extracts the acoustic BPF and a direction-of-arrival (DOA) estimate from a four-microphone array using multi-channel FFT and GCC-PHAT, and (iii) fuses the two streams with a sequential Bayesian updater that includes harmonic-aware frequency cross-validation. The central finding is methodological: detection *architecture*, not parameter tuning, determines success. A grid-FFT baseline failed at range regardless of tuning, whereas replacing it with per-pixel frequency extraction and connected-component clustering produced stable, zero-false-positive detection on our recordings, with first-detection latency reduced from 600–1200 ms to approximately 100 ms after real-time optimization. Acoustic BPF detection and cross-modal frequency agreement worked reliably; cross-modal *spatial* (azimuth) calibration did not converge to a usable model under our indoor test conditions ($R^2 = 0.137$), which we report as an instructive negative result.**

***Index Terms*—event camera, neuromorphic vision, drone detection, counter-UAS, sensor fusion, microphone array, blade-pass frequency, Bayesian inference**

I. Introduction

Unauthorized consumer drones have moved from nuisance to operational threat. The 2018 Gatwick Airport incident diverted roughly 1,000 flights, stranded 140,000 passengers, and caused losses estimated at \$65M; drone-borne contraband smuggling into correctional facilities has grown by an order of magnitude in recent years, and hundreds of illegal incursions near airports are now reported per quarter. Consumer platforms are inexpensive (\$300–\$2,000), widely available, and—critically for detection—increasingly autonomous: a drone flying a pre-programmed GPS waypoint mission emits no radio signal at all, rendering it invisible to RF scanning, the dominant commercial detection technology. The legal environment amplifies the need for *passive* detection specifically: in the United States only four federal agencies may lawfully jam or neutralize a drone, so airports, prisons, police, and private security are restricted to detection alone.

TABLE I
Structural Weaknesses of Existing Detection Modalities

Modality	**Mechanism**	**Structural weakness**
RF scanning	Detect drone radio protocol	Blind to autonomous / RF-silent drones
Radar	Reflected RF	Cannot separate small drones from birds; high cost
RGB / thermal	Frame imaging (30–60 fps)	Motion blur, $\sim$60 dB dynamic range, fails in low light / fog
Acoustic only	Propeller sound	Short range; 10–20 % false-positive rate in noise

Each existing modality has a structural blind spot, summarized in Table I. Radar offers kilometer-scale range but cannot reliably separate small drones from birds of similar radar cross-section (0.01–$0.1\,\mathrm{m}^2$). Frame-based cameras integrate over 16–33 ms exposures: a propeller at 5,000 RPM completes a revolution in 12 ms and appears only as blur, and $\sim$60 dB of dynamic range fails at dawn, dusk, fog, and backlit sky. Acoustic-only systems reach 100–150 m in quiet environments but degrade rapidly in noise, with false-positive rates of 10–20 % because HVAC, generators, and vehicles overlap the drone frequency band.

A. Research Question and Hypothesis

We ask: can a *passive* sensor combination exploit the physics-guaranteed propeller blade-pass frequency (BPF) to detect drones with a false-alarm rate far below acoustic-only systems, without any RF emission?

Our hypothesis is that because the BPF appears simultaneously in the optical and acoustic domains, requiring *cross-modal frequency agreement* should suppress the dominant false-alarm sources—each of which usually manifests in only one modality (flickering luminaires and displays are silent; HVAC and traffic do not flicker)—driving the joint false-positive probability sharply down.

B. Contributions

This paper makes the following contributions:

1) A working prototype, to our knowledge the first, that fuses a neuromorphic event camera with a microphone array for drone detection via the shared BPF observable.
2) An architectural ablation across nine development phases demonstrating that per-pixel frequency extraction with connected-component spatial clustering succeeds where a grid-FFT baseline fails at range regardless of parameter tuning—a transferable lesson for event-based detection pipelines.
3) A harmonic-aware cross-modal frequency validation scheme and a gated sequential Bayesian fusion framework, with observed accumulate/decay dynamics matching design intent.
4) Real-time engineering techniques (downscaled morphology, pre-allocated buffers, region-of-interest statistics, quality-weighted Bayesian confirmation) that reduce first-detection latency by 6–12$\times$ to $\sim$100 ms.
5) An honest account of negative results, including the failure of visual-to-acoustic azimuth calibration to converge indoors, and the unsuitability of a speech-oriented DOA framework (ODAS) for impulsive propeller acoustics.

II. Related Work

Event-based vision. Neuromorphic event cameras descend from the Dynamic Vision Sensor of Lichtsteiner *et al.* [1]: each pixel independently compares the current log-intensity to a stored reference and asynchronously emits an event $(x, y, t, \mathrm{polarity})$ when the difference crosses a threshold, mimicking retinal ganglion cells [2]. Later sensors combined events with frames [3], and the field is comprehensively surveyed by Gallego *et al.* [4], covering optical flow, tracking, SLAM, and vibration monitoring—but not drone or propeller detection. Event-based moving-object detection and motion segmentation have been demonstrated [6], [7], again without acoustic fusion. Modern sensors such as the Sony IMX636 provide HD resolution in a stacked back-illuminated process [5].

Counter-UAS detection. Surveys of drone detection cover RF, radar, electro-optical, and acoustic approaches [8], [9]. RF fingerprinting achieves strong classification of emitting drones [10] but is by construction blind to RF-silent targets. Acoustic drone detection via the propeller's harmonic signature is well established [11], [12], including microphone-array DOA localization [13], but suffers short range and high false-alarm rates in noise. We are not aware of prior work fusing an event camera with a microphone array for counter-UAS purposes—or, indeed, for any detection application.

Acoustic DOA. Generalized cross-correlation with phase transform (GCC-PHAT) [14] remains the standard lightweight time-difference-of-arrival estimator; ODAS [15] provides an integrated localization/tracking stack tuned primarily for speech.

Light-flicker positioning. A companion line of work uses luminaire flicker as a location fingerprint [16]–[18]; we discuss a preliminary feasibility study in Section XI.

TABLE II
Conventional Frame Camera vs. Event Camera

Property	Frame camera	Event camera	Factor
Temporal resolution	33 ms (30 fps)	$\sim$1 µs	3×10^4
Dynamic range	$\sim$60 dB	$>$120 dB	10^3 (dB ratio)
Motion blur	severe	none	—
Latency	$\geq$33 ms	$<$100 µs	$\sim$330
Power	1–5 W	5–205 mW	10–100
Data rate	constant, high	motion-proportional	10–1000
Low light	fails $<$1 lux	$\sim$80 mlux	$\sim$12

III. Background: Why an Event Camera

Table II contrasts the two imaging paradigms. The decisive property for this application is temporal resolution. A drone BPF of 200 Hz has a 5 ms period; a 30 fps camera (33 ms/frame) captures well under one sample per period—far below the Nyquist rate—whereas an event camera with microsecond timestamps samples thousands of times per period, permitting direct per-pixel frequency measurement. Because each pixel fires independently with no exposure integration, there is no motion blur: a blade is registered at its exact position at the microsecond it crosses a pixel. The $>$120 dB dynamic range spans moonless night ($\sim$80 mlux) to direct sunlight (100 klux), and the data rate is proportional to scene activity: a sensor watching empty sky emits almost nothing until a drone enters the field of view.

IV. Physical Principle

A propeller with B blades rotating at shaft rate f_s (Hz) modulates both reflected light and radiated sound at the blade-pass frequency

$$f_{\mathrm{BPF}} = B \cdot f_s = \frac{B \cdot \mathrm{RPM}}{60}. \tag{1}$$

Optically, each blade sweep changes local scene luminance, generating events at f_{BPF} and its harmonics; acoustically, each blade sheds a pressure wave at the same fundamental. The signature is unavoidable (a multi-rotor must spin its propellers to fly), narrowband, and rare in nature: very few open-air phenomena produce stable narrowband oscillations in the 100–500 Hz band. Table III lists hover-regime BPF ranges for common platforms; racing quadrotors extend the band to 1–2.5 kHz. Because the two channels share a single physical cause, requiring their measured frequencies to agree (up to harmonic ambiguity, Section VII) rejects noise sources that are periodic in only one modality—flickering luminaires, wind noise, speech, or vibrating structures.

V. Visual Detection Pipeline

The visual pipeline went through nine documented development phases; the crucial step was an architectural replacement, not tuning.

TABLE III
TYPICAL PROPELLER SIGNATURES OF COMMON DRONES (HOVER)

Platform	Prop	RPM	BPF
DJI Mini 3 Pro	4×2-blade, 16 cm	7000–9000	233–300 Hz
DJI Mavic 3	4×2-blade, 21 cm	5000–7500	167–250 Hz
DJI Phantom 4	4×2-blade, 24 cm	4500–6500	150–217 Hz
DJI Inspire 2	4×2-blade, 38 cm	3000–5000	100–167 Hz
DJI Matrice 600	6×2-blade, 53 cm	2500–4000	83–133 Hz
Racing quad	4×3-blade, 13 cm	20k–40k	1–2 kHz

A. Baseline: Grid-Based FFT (v1) — Failure at Range

The first detector divided the 1280×720 sensor into a 16×16 grid, counted events per cell per $500\,\mu$s slice, ran a rolling-window FFT on each cell's event-rate time series, thresholded on SNR, and clustered adjacent active cells. It ran correctly over $\sim$175 s of recorded data but detected the propeller only at a range of a few centimeters. Root-cause analysis identified three fatal flaws: (1) *no spatial coherence*—a 160×90-pixel cell mixes a distant propeller's few pixels with hundreds of noise pixels, diluting the signal below the cell noise floor; (2) *no temporal persistence*—each frame was analyzed independently, so noise spikes passing the SNR threshold were reported as confidently as real propellers; and (3) *fixed grid boundaries*—a propeller straddling a cell boundary splits its signal across 2–4 cells, halving or quartering the effective SNR.

B. Parameter Tuning — A Key Negative Result

To chase range, thresholds were relaxed aggressively (filter length $7 \rightarrow 4$, maximum period difference $500 \rightarrow 1500\,\mu$s, minimum pixel count $25 \rightarrow 5$, grid $16 \rightarrow 8$ cells, FFT window $0.5 \rightarrow 1.0$ s, minimum SNR $3.0 \rightarrow 2.0$). The result was 25–43 "propeller regions" per frame scattered across random frequencies—pure noise promoted to false positives. *Parameter tuning cannot repair a fundamentally wrong architecture.*

C. Per-Pixel Frequency + Connected Components (v2)

The redesigned pipeline is

$$\begin{aligned}\text{per-pixel freq. map} &\rightarrow \text{binary mask} \rightarrow \text{dilation}\\ &\rightarrow \text{connected components}\\ &\rightarrow \text{freq.-consistency filter} \rightarrow \text{tracking.}\end{aligned}$$

The SDK's asynchronous frequency-map algorithm already reports, for every pixel, the dominant frequency of brightness oscillation—the correct foundation, free of grid quantization. Three design choices proved decisive:

- **Spatial clustering instead of a fixed grid.** Connected-component labeling adapts to the propeller's true shape; even 2–3 connected pixels at a consistent frequency are meaningful, while the same pixels scattered across a grid cell are not. Only actually-vibrating pixels are examined, so there is no signal dilution.
- **Frequency-consistency gate.** Within each cluster the coefficient of variation of per-pixel frequency must satisfy $\mathrm{CV} \leq 0.3$.
- **Temporal tracking.** Greedy association on spatial proximity (<100 px) and frequency similarity ($<30\,\%$); confirmation only after a minimum number of consecutive hits; pruning after 8 idle frames; exponential smoothing ($\alpha = 0.3$) of position and frequency.

TABLE IV
V2 CALLBACK COST BREAKDOWN (FULL RESOLUTION, 25 HZ)

Operation	Per call	At 25 Hz
Dilation (11×11, 921 kpx)	3–5 ms	$\sim$100 ms/s
Connected components (921 kpx)	5–10 ms	$\sim$175 ms/s
Frame copy (2.7 MB)	1–2 ms	$\sim$40 ms/s
Total	$\sim$15 ms	$\sim$315 ms/s

The grid analyzer and its clustering were deleted outright. On the bench propeller this yielded a single stable track at $\sim$99.7 Hz, locked position, $\sim$100 active pixels, and 270+ consecutive hits over the recording, with transient few-pixel false positives pruned within 1–2 s.

D. Blade-Count Correction

The test propeller later proved to be 3-bladed: a detected 270 Hz tone is the blade-pass frequency, not the shaft rate, giving $270/3 \times 60 = 5{,}400$ RPM by (1). The search band was accordingly widened to 300 Hz and the blade count set to 3 for correct RPM reporting—an instance of the harmonic ambiguity revisited in Section VII.

E. Ablation: Minimum Cluster Size

Hypothesizing that a distant propeller might excite only one pixel, we reduced the minimum cluster size to a single pixel. This produced 13–17 "confirmed propellers" per frame at random frequencies (19–233 Hz): any noise spike persisting long enough became a detection. The experiment confirms that *spatial coherence ($\geq$2 connected pixels) is the primary false-positive filter*—a single pixel carries no spatial information and is indistinguishable from sensor noise. The final configuration (minimum 2 pixels, 5 confirmation hits) yielded clean detections with zero false positives and multi-pixel clusters (35–269 px) tracked over 80+ hits in a 32 s recording.

F. Real-Time Optimization

The v2 detector was functionally correct but operationally unusable: the display fell many seconds behind by the end of a 175 s recording. The synchronous frequency-map callback ran full-resolution morphology, connected-component analysis, and a frame copy at 25 Hz; Table IV shows the measured cost breakdown, totaling $\sim$315 ms of CPU per wall-clock second and leaving too little time for the 2,000 event slices per second.

Four fixes eliminated the drift: (1) $4\times$ downscaling ($1280 \times 720 \rightarrow 320 \times 180$, a $16\times$ pixel reduction) before morphology, with frequency statistics still computed on full-resolution regions of interest; (2) reducing the update rate to 10 Hz; (3) in-place frame drawing, eliminating $\sim$67 MB/s of copying;

TABLE V
FRAMES TO 95 % CONFIDENCE VS. DETECTION QUALITY (20 HZ)

Quality	p per obs.	Frames (time)
Weak (2 px, CV = 0.25)	0.67	4 (200 ms)
Medium (20 px, CV = 0.15)	0.74	3 (150 ms)
Strong (100+ px, CV = 0.05)	0.83	2 (100 ms)

TABLE VI
VISUAL DETECTOR PERFORMANCE, V2 VS. V3

Metric	v2	v3	Gain
First detect (strong signal)	600–1200 ms	∼100 ms	6–12×
First detect (weak signal)	600–1200 ms	∼200 ms	3–6×
Steady-state update	100–140 ms	50–90 ms	∼2×
Per-frame analysis	∼1.5 ms	∼0.8 ms	∼2×
GC jitter	±5 ms	$< \pm 1$ ms	∼5×

and (4) early exit on near-empty masks. Callback cost dropped to ∼20 ms/s with zero latency drift over the full recording.

A production pass (v3) then targeted first-detection latency. Buffers are pre-allocated once and reused, eliminating ∼20 MB/s of per-frame allocation and the resulting garbage-collection jitter; tracking is velocity-predictive, matching detections against each track's predicted position; and the update rate was doubled to 20 Hz. Most importantly, the hard hit-count confirmation (5 hits at 10 Hz = 500 ms minimum) was replaced by quality-weighted Bayesian confidence: each observation contributes likelihood

$$p = 0.65 + q\,(0.85 - 0.65), \tag{2}$$

where $q \in [0, 1]$ combines a pixel-count score (ramping over 1–50 pixels) and a frequency-consistency score (ramping over CV = 0.3 → 0), accumulated over n observations as

$$P = 1 - (1 - p)^n \tag{3}$$

and decayed by $0.8\times$ per idle frame, with confirmation at $P \geq 0.95$. Unlike a hard threshold, which treats a 2-pixel cluster at CV = 0.29 identically to a 200-pixel cluster at CV = 0.01, this lets strong detections confirm in 2 frames while weak ones take 4 (Table V), preserving the zero-false-positive behavior. Table VI summarizes the measured improvement.

VI. ACOUSTIC DETECTION PIPELINE

A. BPF Extraction

A rolling 0.2 s window (3200 samples at 16 kHz) per channel is Hann-windowed and transformed; magnitude spectra are incoherently averaged across the four channels (∼6 dB SNR gain). The peak is picked in the target band, its SNR computed against the median noise floor, and detections below a minimum SNR of 3.0 rejected. This proved reliable on sustained propeller tones; source calibration on the bench propeller measured a dominant tone at 162.7 Hz with SNR ≈ 24 dB.

B. DOA via GCC-PHAT

For each opposing microphone pair (64 mm baseline) we compute the phase-transform cross-correlation

$$R_{12}(\tau) = \int \frac{X_1(\omega)X_2^*(\omega)}{|X_1(\omega)X_2^*(\omega)|}\, W(\omega)\, e^{j\omega\tau}\, d\omega, \tag{4}$$

with a frequency weighting $W(\omega)$ emphasizing the target band, $4\times$ zero-padded inverse transform for sub-sample delay resolution, parabolic peak interpolation, and a peak-to-second-peak confidence-ratio gate (≥ 1.5). Pair estimates are fused by averaging when both are confident, followed by a 7-sample median filter; the sample spread gates unstable bearings. This yields usable azimuth on steady tones; the planar 32 mm-radius array cannot resolve elevation and provides $\sim \pm 10^\circ$ azimuth accuracy at range.

C. ODAS — A Negative Result

We integrated the ODAS framework [15] via a custom socket relay. It produced robust DOA on speech (sustained utterances of 500 ms or more) but was unreliable on propeller acoustics, which are comparatively impulsive and broadband. This motivated the custom GCC-PHAT implementation above. The array's on-board DSP additionally exposes a firmware DOA with speech-tuned voice-activity detection; it was likewise unreliable on steady tones and was not fused.

VII. SENSOR FUSION

A. Harmonic-Aware Frequency Cross-Validation

One modality frequently reports the blade-pass tone while the other reports the shaft fundamental or a harmonic. We therefore search harmonic pairs (n_v, n_a), $1 \leq n_v, n_a \leq 4$, and declare a match if

$$\frac{|f_v/n_v - f_a/n_a|}{f_a/n_a} < 0.05, \tag{5}$$

returning confidence $1/(n_v n_a)$ so that fundamental-to-fundamental agreement outweighs high-order coincidences.

B. Sequential Bayesian Updating

We chose sequential Bayesian updating over a rule-based conjunction ("camera AND microphone ⇒ alarm") because it accumulates graded evidence: a weak detection adds a small increment, a cross-modally corroborated one a large increment. Starting from prior $P(D) = 0.001$, each modality contributes a likelihood-ratio update with sigmoid likelihoods:

$$P(V \mid D) = \sigma(\mathrm{SNR}_v - 5), \quad P(V \mid \neg D) = 0.01\, e^{-0.5\,\mathrm{SNR}_v}, \tag{6}$$

$$P(A \mid D) = \sigma(\mathrm{SNR}_a - 4), \quad P(A \mid \neg D) = 0.05, \tag{7}$$

with the acoustic false-alarm likelihood reduced to 0.005 for elevated sources, since few non-drone sources radiate from above ∼10° elevation. A cross-modal frequency match contributes $P(F \mid D) = 0.99 \cdot \mathrm{conf}$ against $P(F \mid \neg D) = 0.001$. When no evidence arrives, the posterior decays toward the prior by $0.995\times$ per frame. A drone is declared at $P > 0.8$.

C. Fusion Gating

To prevent overconfident fusion from coincidental evidence, several gates proved necessary in practice: a visual quality gate (minimum pixel count and track confidence ≥ 0.98); an acoustic stability gate (persistence and frequency jitter $\leq 25\,\mathrm{Hz}$); a temporal association gate ($|t_v - t_a| \leq \sim 200\,\mathrm{ms}$); an angle-residual gate (visual azimuth vs. acoustic DOA within 35°); and stricter audio-only gates (SNR $\geq 6\,\mathrm{dB}$ *and* bounded DOA spread) before an audio-only update may raise $P(D)$. In a live run with no propeller in view, the detector repeatedly latched $P(D) \to \sim 1.0$ on strong 140–200 Hz tones through the audio-only path and decayed back when the source stopped—the accumulate/decay dynamics behaved as designed, and the run illustrates that single-modality dominance is a real failure mode that the gates must police.

VIII. Implementation

Visual. Prophesee EVK4-HD evaluation kit with the Sony IMX636 sensor [5]: 1280×720 resolution, 4.86 µm pixels, $<$100 µs pixel latency at 1000 lux, $>$120 dB dynamic range, 5 mW standby / 205 mW maximum power. Optics: a 220° fisheye lens for hemispheric sky coverage ($\sim$5.8 px/degree) and a rectilinear C-mount lens (63.8° HFOV) for azimuth experiments; Metavision SDK frequency-map algorithms. **Acoustic.** ReSpeaker 4-mic circular USB array (32 mm radius, 64 mm opposing-pair baseline), 16 kHz / 16-bit capture on a background thread; on-board XMOS XVF-3000 DSP and a 12-LED ring used as a physical bearing indicator. Pixel-to-azimuth mapping uses a pinhole model, $f_{\mathrm{px}} = (W/2)/\tan(\mathrm{HFOV}/2)$, $\mathrm{az} = \mathrm{atan2}(x - W/2,\, f_{\mathrm{px}})$. **Software.** Python 3.9 (required by the SDK binaries), NumPy, OpenCV, PyAudio. The visual Nyquist constraint is $f_{\max} \leq 10^6/(2\,\Delta t)$ for accumulation period Δt in µs (1 kHz at the default $\Delta t = 500\,\mu\mathrm{s}$). Session traces are logged as JSONL with per-interval posterior, evidence, association lag, and gate states. A synthetic on-screen vibration generator (shapes flickering at 15–120 Hz) validates the visual frequency pipeline end-to-end without a live drone, and an event-centroid handshake analyzer validates low-frequency (3–8 Hz) response and characterizes platform vibration for future stabilization.

IX. Experiments and Results

All experiments used a bench propeller as a controlled rotating source and the synthetic vibration generator for pipeline validation. Recordings were indoor, single-source, and $\sim$30–175 s long.

A. Summary of Outcomes

Table VII consolidates what worked and what failed across the study. The headline quantitative results are the latency figures of Table VI and the zero-false-positive tracking behavior of the final visual configuration.

TABLE VII
CONSOLIDATED OUTCOMES

Method / component	Outcome
Per-pixel flicker-frequency extraction	✓
Connected-component spatial clustering	✓
Temporal tracking + Bayesian confidence	✓
Real-time latency engineering	✓
Acoustic BPF via multi-channel FFT	✓
GCC-PHAT azimuth on sustained tones	✓
Harmonic-aware frequency cross-validation	✓
Gated sequential Bayesian fusion	✓
Grid-FFT architecture (v1)	×
Parameter tuning to extend range	×
Single-pixel minimum cluster size	×
ODAS for propeller acoustics	×
Visual→acoustic azimuth calibration	×
Elevation DOA (planar array)	×

TABLE VIII
PROJECTED DETECTION RANGE BY PLATFORM CLASS (DESIGN ESTIMATES, NOT MEASURED)

Platform class	Visual	Acoustic	Fused
Heavy-lift (38 cm props)	24 m	150 m	150 m
Phantom-class (24 cm)	15 m	120 m	120 m
Mavic-class (21 cm)	13 m	100 m	100 m
Mini-class (12 cm)	7 m	50 m	50 m

B. Projected Range and False-Positive Rate

Visual range is limited by angular resolution: the 220° fisheye yields $\sim$5.8 px/degree, so a 21 cm propeller at 15 m subtends $\sim 0.8° \approx 4.6$ pixels—near the minimum for frequency detection. Extrapolating from this geometry and from acoustic range-profiling of the bench source gives the design projections of Table VIII: acoustic sensing extends the detection envelope to $\sim$50–150 m depending on platform class, with visual confirmation and frequency cross-validation inside its shorter range. Similarly, published acoustic-only false-positive rates of 10–20 % and our single-modality behavior suggest that requiring cross-modal frequency agreement should drive the joint false-positive rate below 0.1 %. *We stress that these range and FPR figures are design-derived projections, not measurements*; no outdoor or labeled ground-truth validation has been performed (Section XI).

C. Azimuth Calibration — A Negative Result

To validate the angle-residual gate we attempted to fit a linear map $\mathrm{DOA}_a = g \cdot \mathrm{az}_v + b$ by sweeping a co-located source laterally, binning samples by visual azimuth with robust per-bin medians, and fitting with iterative MAD outlier rejection and an R^2 solidity check (threshold 0.6). The best fit (rectilinear lens, 63.8° HFOV) achieved $g = -0.011$, $b = 0.37°$, $R^2 = 0.137$, RMSE 0.70°—not solid. The acoustic DOA spanned only $\sim \pm 1.2°$ while the visual sweep spanned $\sim \pm 25°$, consistent with indoor reverberation, an insufficiently wide or slow sweep, or low GCC-PHAT confidence during the session. Cross-modal *spatial* fusion therefore remains unvalidated; cross-modal *frequency* fusion is validated.

TABLE IX
LIMITATIONS AND THREATS TO VALIDITY

Area	Limitation
Dataset	Indoor, single source, short recordings
Range / FPR	Projections only; no outdoor / labeled tests
Spatial fusion	Azimuth calibration weak ($R^2 = 0.137$)
DOA	Planar 32 mm array; $\sim \pm 10°$, no elevation
Multi-target	One source per cluster assumed
Synchronization	No hardware clock; $\sim$100 ms uncertainty
Harmonics	Blade-count / shaft-rate ambiguity

X. Discussion and Lessons Learned

Architecture beats tuning. The single most consequential change in the project was architectural: replacing grid-FFT event counting with per-pixel frequency extraction and connected-component clustering. No amount of threshold relaxation rescued the grid detector; it merely promoted the noise floor to false positives.

Spatial coherence is the false-positive filter. The single-pixel ablation demonstrates that requiring as few as two connected pixels at a consistent frequency removes essentially all sensor-noise detections. Persistence (temporal tracking) removes roughly 95 % of the remaining transients.

Probabilistic beats deterministic confirmation. Quality-weighted Bayesian confidence achieves the same false-positive suppression as a hard hit-count threshold while confirming strong detections 3–6× faster, because it does not treat a 200-pixel low-CV cluster and a 2-pixel marginal cluster identically.

Profile before porting. The v2 detector was algorithmically correct but operationally unusable; a cost breakdown (Table IV) showed that a simple 4× downscale solved the problem with no loss in detection quality—well before any C++ rewrite was warranted.

Modality dominance is a real failure mode. The audio-only run in Section VII shows that a Bayesian fuser will happily saturate on one strong modality; explicit quality, association, and angle gates—not the update rule itself—are what make fusion trustworthy.

Tool-domain mismatch. ODAS and the array's firmware DOA, both tuned for speech, failed on propeller acoustics—a reminder that off-the-shelf audition stacks encode source assumptions that must be checked before integration.

XI. Limitations and Future Work

Table IX lists the principal threats to validity. All detection results are from indoor, short, single-source lab recordings; no outdoor, multi-drone, or GPS-ground-truth validation was performed, so the figures of Section IX-B are projections, and the sub-0.1 % false-positive-rate target must be established empirically on a labeled dataset. The 32 mm planar array cannot resolve elevation; harmonic ambiguity (a 2-blade propeller at 100 Hz BPF is indistinguishable from a 4-blade at 50 Hz shaft rate) requires external metadata; and there is no hardware inter-sensor clock ($\sim$100 ms association uncertainty).

Planned engineering work. A C++ port would remove interpreter, GIL, and garbage-collection overheads for deterministic sub-millisecond jitter; headless operation saves a further 8–12 ms per cycle; SIMD thresholding and double-buffered frequency maps would support 50–100 Hz analysis rates. On embedded targets (e.g., Jetson-class modules), offloading per-pixel period detection to FPGA fabric alongside real-time scheduling projects first-detection latency of $\sim$80–100 ms and steady-state updates of $\sim$20 ms. Sensor work includes a non-planar or larger-baseline array for elevation, hardware time synchronization, multi-target tracking, acoustic-guided visual search (using DOA to steer the visual search region), and repeating the azimuth calibration outdoors with a wider, slower sweep.

A companion feasibility study on using ceiling-light flicker frequencies as location fingerprints—inspired by LiTell [16], iLAMP [17], and event-camera visible-light positioning [18]—completed a literature review and utility implementation with a positive feasibility verdict, but end-to-end validation remains open.

XII. Conclusion

We demonstrated a passive, RF-silent drone-detection prototype that fuses neuromorphic vision and acoustic sensing through the shared blade-pass frequency observable—a sensor combination with, to our knowledge, no precedent in the literature. The validated contributions are the architectural comparison (per-pixel frequency plus connected components decisively outperforms grid FFT), the real-time engineering that brings first-detection latency to $\sim$100 ms in Python, quality-weighted Bayesian confirmation, harmonic-aware cross-modal frequency validation, and gated sequential Bayesian fusion whose accumulate/decay dynamics behave as designed. We report the failure of indoor visual-to-acoustic azimuth calibration as a genuine, instructive negative result rather than an omission. Range and false-positive-rate figures remain design projections pending outdoor validation on labeled data.

References

[1] P. Lichtsteiner, C. Posch, and T. Delbruck, "A 128 × 128 120 dB 15 μs latency asynchronous temporal contrast vision sensor," *IEEE J. Solid-State Circuits*, vol. 43, no. 2, pp. 566–576, 2008.

[2] C. Posch, T. Serrano-Gotarredona, B. Linares-Barranco, and T. Delbruck, "Retinomorphic event-based vision sensors: Bioinspired cameras with spiking output," *Proc. IEEE*, vol. 102, no. 10, pp. 1470–1484, 2014.

[3] C. Brandli, R. Berner, M. Yang, S.-C. Liu, and T. Delbruck, "A 240 × 180 130 dB 3 μs latency global shutter spatiotemporal vision sensor," *IEEE J. Solid-State Circuits*, vol. 49, no. 10, pp. 2333–2341, 2014.

[4] G. Gallego *et al.*, "Event-based vision: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 1, pp. 154–180, 2022.

[5] T. Finateu *et al.*, "A 1280 × 720 back-illuminated stacked temporal contrast event-based vision sensor with 4.86 μm pixels, 1.066 GEPS readout, programmable event-rate controller and compressive data-formatting pipeline," in *IEEE Int. Solid-State Circuits Conf. (ISSCC)*, 2020, pp. 112–114.

[6] A. Mitrokhin, C. Fermüller, C. Parameshwara, and Y. Aloimonos, "Event-based moving object detection and tracking," in *IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2018, pp. 1–9.

[7] T. Stoffregen, G. Gallego, T. Drummond, L. Kleeman, and D. Scaramuzza, "Event-based motion segmentation by motion compensation," in *IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2019, pp. 7244–7253.

[8] I. Guvenc, F. Koohifar, S. Singh, M. L. Sichitiu, and D. Matolak, "Detection, tracking, and interdiction for amateur drones," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 75–81, 2018.
[9] B. Taha and A. Shoufan, "Machine learning-based drone detection and classification: State-of-the-art in research," *IEEE Access*, vol. 7, pp. 138669–138682, 2019.
[10] M. Ezuma, F. Erden, C. K. Anjinappa, O. Ozdemir, and I. Guvenc, "Micro-UAV detection and classification from RF fingerprints using machine learning techniques," *IEEE Access*, vol. 8, pp. 7654–7668, 2020.
[11] J. Mezei, V. Fiaska, and A. Molnár, "Drone sound detection," in *IEEE Int. Conf. Cognitive Infocommunications (CogInfoCom)*, 2015, pp. 333–338.
[12] A. Bernardini, F. Mangiatordi, E. Pallotti, and L. Capodiferro, "Drone detection by acoustic signature identification," in *Proc. IS&T Electronic Imaging*, 2017, pp. 60–64.
[13] Z. Shi, X. Chang, C. Yang, Z. Wu, and J. Wu, "An acoustic-based surveillance system for amateur drones detection and localization," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 2731–2739, 2020.
[14] C. Knapp and G. Carter, "The generalized correlation method for estimation of time delay," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 24, no. 4, pp. 320–327, 1976.
[15] F. Grondin and F. Michaud, "Lightweight and optimized sound source localization and tracking methods for open and closed microphone array configurations," *Robot. Auton. Syst.*, vol. 113, pp. 63–80, 2019.
[16] C. Zhang and X. Zhang, "LiTell: Robust indoor localization using unmodified light fixtures," in *Proc. ACM MobiCom*, 2016, pp. 230–242.
[17] S. Zhu and X. Zhang, "Enabling high-precision visible light localization in today's buildings," in *Proc. ACM MobiSys*, 2017, pp. 96–108.
[18] G. Wang *et al.*, "Visible light positioning using an event camera," *IEEE Sensors*, 2020.